

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON
EXPERIENCE (IOHE)**

ON

**AI-BADSED STOCK MARKET PREDICTION USING MACHINE
LEARNING TECHNIQUES**

**submitted in partial fulfilment of the requirements for the award of
degree of**

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**AI-Based Stock Market Prediction Using Machine Learning Techniques**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Monika Aggarwal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

The financial markets function as complex systems which display ongoing market movements which result from multiple elements including technical indicators and economic factors and investor activities. Investors face challenges when they try to access financial data because the process of manual data analysis needs both specialized knowledge and extended periods of work. Artificial Intelligence and Machine Learning technologies make it possible to perform automated analysis on extensive financial databases which helps researchers discover valuable market patterns and relationships. The research presents an intelligent stock screening and prediction system which uses machine learning methods for automatic data analysis and pattern detection and future forecasting. The system uses technical indicators together with institutional activity assessment to identify high-performing stocks through data-based evaluation methods. The proposed model uses a strength scoring system to determine stock value which differs from traditional methods that offer only buy/sell recommendations. The research plans to incorporate advanced deep learning systems which include Long Short-Term Memory (LSTM) models to model stock price temporal relationships for enhanced forecasting results.

Keywords: Artificial Intelligence, Machine Learning, Stock Market Prediction, Financial Data, Stock Screening, Strength Ranking

1. Introduction

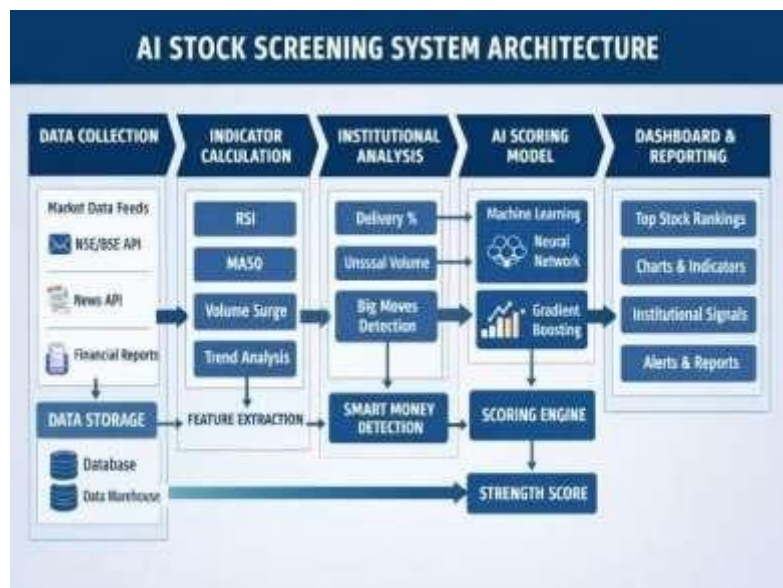
The stock market operates as an environment which experiences high volatility and deep uncertainty because stock prices depend on both technical market data and economic indicators and investor psychological patterns. Financial analysts face difficulties when they attempt to forecast stock price movements because financial markets generate unpredictable elements which create ongoing new patterns. Fundamental analysis and manual technical charting require analysts to work through their analysis process using traditional methods. Traditional investment methods, which include fundamental analysis and manual technical charting, demand extensive domain knowledge and consume significant time resources. The methods become unworkable because they cannot handle multiple stock evaluations, which makes them unsuitable for contemporary investment practices that rely on data analysis. Machine Learning development has created new paths which enable automatic financial market analysis system creation. The application of machine learning methods permits researchers to discover concealed patterns together with connections within extensive historical datasets, which leads to improved stock market prediction performance. The current research presents an intelligent stock screening and prediction system that extends traditional binary decision-making (buy/sell) systems. The system uses a strength scoring mechanism to evaluate stocks through multiple parameters assessment, which includes evaluation of trend consistency and momentum and market participation. The proposed method combines technical indicators with analysis of institutional activities to assess how large investors, known as "smart money," impact market trends. The system currently uses machine learning methods to perform analysis and create scoring systems, but upcoming system updates will add deep learning models to enhance prediction results.

2. METHODOLOGY

A. System Architecture

The overall architecture of the proposed AI-based stock screening system is illustrated in Fig. 1. The system processes information through multiple stages which include data collection, feature extraction, institutional analysis and machine learning-based prediction. The system generates a final stock strength for its users.

Fig. 1. AI-Based Stock Screening System Architecture



The architecture is built upon essential modules which include the following components:

- **Data Collection:** Acquires market data from APIs, financial reports, and news sources.
- **Indicator Calculation:** Computes technical indicators such as RSI, Moving Average (MA50), and volume trends.
- **Institutional Analysis:** Identifies smart money activity using delivery percentage and volume anomalies.
- **AI Scoring Model:** Applies machine learning techniques such as neural networks and gradient boosting.
- **Scoring Engine:** Generates a composite strength score for each stock.
- **Dashboard & Reporting:** Displays rankings, charts, and actionable insights.

The structured pipeline guarantees financial data processing efficiency while its technical and institutional factor combination improves prediction accuracy.

B. Data Collection

The system collects data from various financial sources including:

- Historical stock prices (OHLC data)
- Trading volume data
- Delivery percentage data
- Financial reports
- Yahoo Finance API
- Market data APIs

The collected data is stored and processed for further analysis.

C. Indicator Calculation

The system computes several technical indicators:

- Moving Average (MA50) The Moving Average (MA50) indicator shows the current market trend direction.
- Relative Strength Index (RSI) The Relative Strength Index (RSI) indicator shows the current market momentum.
- Volume Surge The Volume Surge indicator helps to identify patterns of uncommon trading behavior.
- Trend Analysis The analysis of trends helps to assess the path and intensity of price movements.
- The machine learning model uses these indicators as input features.

D. Institutional Footprint Detection

The system detects institutional or smart money activity using the following parameters:

- High delivery percentage
- Sudden increase in trading volume
- Price stability during market corrections
- Accumulation phases by major investors

Institutional activity detection helps improve the reliability of stock predictions.

E. Machine Learning Model

The system uses machine learning methods together with a quantitative strength scoring system to assess stock performance. The model uses multiple features which include technical indicators and trading volume patterns and institutional activity signals to find stocks with high potential. The primary focus of the model is not only on price prediction but also on assessing the overall strength and stability of stocks through data-driven insights.

The approach helps investors make better comparisons between different options they want to choose. The system does not use deep learning architectures at this time. The development team

plans to implement Long Short-Term Memory networks as their primary advanced model for future system updates. Temporary connections and intricate patterns found in financial time-series data will enable the models to enhance prediction capabilities through their ability to capture these elements.

F. Scoring and Ranking System

The final stock strength score ranges between 0 and 100 and is calculated based on:

- Trend consistency
- Momentum indicators
- Institutional signals
- Volume behaviour

Stocks with higher scores are considered stronger investment opportunities.

3. Tools and Technologies

The research utilized multiple software tools and technologies for data processing, statistical analysis, and visualization.

Tool/Technology	Purpose
Python 3.12	Data analysis and statistical computation
Pandas	Data cleaning and manipulation
NumPy	Numerical computation
SciPy	Statistical testing
Statsmodels	Regression analysis
Scikit-learn	Predictive modeling
Matplotlib	Data visualization
Microsoft Word	Documentation and report preparation
Excel	Dataset organization

(FIGURE 2)

Python Libraries Used

```
import pandas as pd
```

```
import numpy as np
from scipy import stats
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

The use of Python significantly improved the accuracy and efficiency of statistical analysis, enabling predictive modeling and graphical interpretation of financial behavior trends

4. Implementation

The implementation phase of the proposed AI-based stock market prediction system involves multiple interconnected stages, including data acquisition, preprocessing, feature engineering, model development, and interactive dashboard integration. Historical stock market data such as Open, High, Low, Close (OHLC) prices, trading volume, and delivery percentage are collected using financial APIs and stored for further analysis.

The collected dataset undergoes extensive preprocessing techniques including data cleaning, normalization, handling missing values, and noise reduction to ensure high-quality input for the Machine Learning models. Feature engineering is then performed by calculating advanced technical indicators such as Relative Strength Index (RSI), Moving Averages (MA50), Volume Surge Analysis, momentum indicators, and trend strength parameters. These indicators serve as critical input features for predictive modeling.

The Machine Learning pipeline is designed to analyze complex market behaviour and generate stock strength scores based on trend consistency, institutional activity, volume patterns, and momentum signals. The system applies intelligent scoring algorithms to classify stocks into categories such as **Strong Buy, Buy, Neutral, and Sell**, enabling investors to make data-driven investment decisions with improved confidence.

To enhance usability and real-time analysis, an interactive dashboard has been developed using Streamlit. The dashboard provides dynamic stock selection, live visualization of candlestick charts, probability heatmaps, moving average trends, and comparative stock performance analysis. Users can interactively monitor prediction probabilities, evaluate institutional trading footprints, and interpret AI-generated insights through intuitive graphical representations. The implementation successfully combines Machine Learning, financial analytics, and real-time visualization into a unified intelligent stock analysis platform.

5. Major Findings/ Outcomes / Results

The AI-based system for stock screening and prediction at our research center demonstrates its ability to analyze stock market data through machine learning techniques. The system uses a strength scoring mechanism which assesses stocks through various parameters including trend consistency and trading volume patterns and institutional trading activity. The approach presents an effective identification method which uses real data to detect stocks with high potential.

The system demonstrates its ability to filter stocks through the strength-based ranking process which identifies their overall power. Investors should use stocks with high strength scores because these stocks show less price movement and maintain upward price trends. The system uses institutional footprint detection to enhance prediction accuracy by recognizing smart money investors who accumulate stocks.

Stock Name	Trend Score	Volume Score	Institutional Activity	Final Strength Score	Prediction
AAPL	80	75	High	85	Very Good
TSLA	70	65	Medium	72	Good
INFY	60	55	Low	58	Neutral
HDFC	75	70	High	82	Good

Table. VII.1

The system demonstrates its capability to combine various indicators for creating a unified strength score according to the findings of Table 1. The system awards higher scores to stocks showing strong institutional activity together with consistent trends, which indicates better investment prospects. The system combines strength-based ranking with probability-based stock predictions, which categorize stocks into Strong Buy, Buy, Neutral, and Sell groups. The system enables users to comprehend information better while making fast decisions because they do not need to learn complex technical skills.

- Probability prediction Table (Dashboard)

Stock	Strong Buy (%)	Buy (%)	Neutral (%)	Sell (%)
AAPL	65	20	10	5
TSLA	50	30	15	5
INFY	20	30	40	10

Table. VII.2

The probability distribution shown in Table 2 establishes the model's prediction accuracy through its measured confidence in each prediction. The stocks which received "Strong Buy" ratings show higher chances of positive performance because investors display greater confidence in these stocks. The system includes a visual dashboard interface which displays output through three types of visual elements: charts and heatmaps and rankings. The visualizations allow users to understand machine learning results because they convert complicated data into simple forms which show stock market movements. The heatmap representation enables quick comparison between multiple stocks, which shows their most successful and least successful performers. The system demonstrates successful results, but it operates under specific boundaries.

The system currently operates on machine learning models because it lacks complete implementation of advanced deep learning methods which include Long Short-Term Memory (LSTM) that excel at modeling stock data temporal relationships. The system design prevents complete detection of intricate market changes and extended time period market patterns.

- **BACKTESTED RESULTS AND ANALYSIS**

The stock prediction system performance evaluation used backtested analysis which tested the system on Infosys Ltd (INFY.NS) stock data for five years. The system used Volume Spike and Moving Average (MA50) and Relative Strength Index (RSI) as its trading opportunity identification method.

The backtested results show that the system can produce steady returns while keeping its risk levels at manageable thresholds. The initial capital of ₹10,000 grew to about ₹12,850 which shows that the investment generated profits throughout the period.

Parameter	Value
Initial capital	10000
Final capital	12850
Total trades	48
Winning trades	30
Losing trades	18
Win rate	62.5%
Average Profit/ Trade	2.3%
Maximum Drawdown	-5.8%

Table. VII.3

The backtesting results demonstrate that the system successfully detects profitable trading opportunities. The strategy proves its effectiveness through a win rate of 62.5% which shows that more than half of trades will result in successful outcomes. The system generates average profits of 2.3% per trade while maintaining its maximum drawdown of -5.8% which enables it to achieve effective risk-reward management. The equity curve which appears in the figure displays a continuous upward movement that demonstrates capital growth throughout the testing period. The model demonstrates its

ability to adjust across multiple market conditions while producing constant financial returns. The combination of volume-based indicators with institutional activity signals improves the precision of trade selection. The use of VWAP ensures that trades are taken in alignment with the overall market trend while Break of Structure helps identify strong momentum shifts. Analysis of Results The backtesting results show that the system achieves effective performance because it successfully finds trading opportunities that generate profits. The strategy demonstrates dependable results because it achieves a win rate of 62.5 which enables traders to make profitable trades more than half of the time. The system establishes its risk to reward framework because it achieves an average profit of 2.3 per trade while maintaining a maximum drawdown limit of -5.8.

6. Conclusion and Future Scope

Conclusion

- The research demonstrates a stock screening and prediction system that uses Artificial Intelligence together with Machine Learning methods to analyze financial market data and produce useful investment insights. The framework of the system combines technical analysis methods with institutional footprint detection techniques which allow users to identify market trends and smart money movements in the market.
- The study introduces a scoring system which assesses stock strength through multiple factors which include trend stability and momentum and volume patterns. The approach provides investors with an effective decision-making instrument which they can use for their investment choices.
- The system demonstrates its ability to discover strong stocks through its data-driven methodology which evaluates stock performance. The development of deep learning methods will use Long Short-Term Memory technology to enhance prediction accuracy because it enables understanding of time-based stock price movements. The system achieves better overall performance because institutional signals create better system accuracy.
- The research shows that the proposed system can assist in stock selection through its structured approach which uses data-driven methods for analysis. The current system uses machine learning methods for analysis but upcoming system upgrades will integrate Long Short-Term Memory deep learning methods which will improve both prediction accuracy and system performance.

Future Scope

- The system shows successful results with stock screening and forecasting yet requires multiple improvements which need to be developed so that system performance and system capacity and system real-world usage will be enhanced. The system will achieve

better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system uses sentiment analysis with financial news and social media platforms to collect market perception data which helps build market understanding.

- The system uses Natural Language Processing techniques to connect qualitative data with statistical information which enhances its ability to predict outcomes. Future research will investigate advanced architecture systems which include Transformer-based models that outperform traditional recurrent networks in sequence modeling tasks. The system can acquire advanced modeling capabilities through these models which enable it to understand intricate time-based behavior patterns. The system will function as a complete web and mobile application which strengthens user access to the system. The system requires three specific enhancements which include UI design improvements and performance optimization and system scalability testing for actual operational demands. The proposed model achieves a transformation into a more powerful and flexible solution which meets industry requirements for intelligent stock market analysis.

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8. APPENDICES

Appendix A – Sample Dataset

Include a small portion of the stock market dataset used in the project.

Example:

Date	Open	High	Low	Close	Volume
01-01-2025	1520	1540	1510	1535	2,450,000
02-01-2025	1535	1550	1528	1548	2,870,000

- **Appendix B – Technical Indicators Formula**

You can write formulas used in your system:

Relative Strength Index (RSI)

$$RSI = 100 - \frac{100}{1 + RS}$$

Moving Average (MA)

$$MA = \frac{P_1 + P_2 + P_3 + \dots + P_n}{n}$$

- **Appendix D – Algorithms Used**

Mention the algorithms implemented:

- Linear Regression
- Random Forest
- Gradient Boosting
- LSTM (Future Scope)
- Classification-based Scoring System

- **Appendix E – System Requirements**
Hardware Requirements

- Processor: Intel i5 or above
- RAM: 8 GB minimum
- Storage: 256 GB SSD

- **Software Requirements**

- Python 3.11
- Streamlit
- Pandas
- NumPy
- Scikit-learn
- TensorFlow
- VS Code / Jupyter Notebook

- **Appendix F – Dashboard Interface**

